**University of Florida, Department of Mathematics**

# Algebra PhD Exam, September 1988

*Do seven problems, at least two from each section.*

## I. Group Theory.

**1.** Let $G$ be the group of all $3 \times 3$ non-singular matrices of determinant 1 with coefficients in the field of 3 elements.

(a) Find $|G|$.

(b) Find a Sylow 3-subgroup of $G$.

(c) Compute $|\operatorname{Syl}_3(G)|$, where $\operatorname{Syl}_3(G)$ is the set of all Sylow 3-subgroups of $G$.

**2.** Let $G$ be a finite group acting transitively on a set $X$, $x \in X$, and $P$ be a Sylow $p$-subgroup of $G_x$ for some prime $p$. Consider the set $\operatorname{Fix}(P) = \{y \in X \mid y^P = y\}$, the set of fixed points of $P$.

(a) Prove that $N_G(P)$ acts on $\operatorname{Fix}(P)$, i.e., maps $\operatorname{Fix}(P)$ to $\operatorname{Fix}(P)$.

(b) Prove that $N_G(P)$ acts transitively on $\operatorname{Fix}(P)$.

**3.** Two permutation representations $G \xrightarrow{\pi_i} \Sigma(X_i)$, (the symmetric group on $X_i$ are equivalent if there exists $\alpha : X_1 \to X_2$ such that for all $g \in G$, $x \in X_1$, we have $x(g\pi_1)\alpha = (x\alpha)(g\pi_2)$. Prove that the following two permutation representations of $S_4$ of degree 12 are not equivalent:

(a) on the right coset space of $\langle(12)\rangle$.

(b) on the right coset space of $\langle(12)(34)\rangle$.

**4.** Let $G$ be a finite group and $A \leq \operatorname{Aut}(G)$ such that $G = [G, A]$. Suppose $N \leq G$ satisfying $[N, A] = 1$. Use the 3-subgroup lemma to prove $N \leq Z(G)$, the center of $G$. (Recall the 3-subgroups lemma: For three subgroups $X, Y, Z$ of a group $M$, if $[X, Y, Z] = 1$ and $[Y, Z, X] = 1$, then $[Z, X, Y] = 1$.)

## II. Homological algebra.

**1.** Let $R$ be a ring and consider the following commutative diagram of $R$-modules and $R$-module homomorphisms such that each row is a short exact sequence.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C & \longrightarrow & 0 \\
 & & {\scriptstyle\alpha}\big\downarrow & & {\scriptstyle\beta}\big\downarrow & & {\scriptstyle\gamma}\big\downarrow & & \\
0 & \longrightarrow & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \longrightarrow & 0
\end{array}$$

Prove that if $\alpha$ and $\gamma$ are isomorphisms then $\beta$ is also an isomorphism.

**2.** Prove the $\mathbb{Z}$-module isomorphisms, where $c = (m, n)$ is the greatest common divisor.

(a) $\mathbb{Z}/m\mathbb{Z} \otimes \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/c\mathbb{Z}$.

(b) $\operatorname{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/c\mathbb{Z}$.

**3.** A module $P$ over a ring $R$ is said to be projective if, given any diagram of $R$-module homomorphisms with $g$ surjective

$$\begin{tikzcd} & P \arrow[d, "f"] \\ A \arrow[r, "g"] & B \end{tikzcd}$$

there exists an $R$-module homomorphism $h : P \longrightarrow A$ such that the following diagram is commutative

$$\begin{tikzcd} & P \arrow[d, "f"] \arrow[dl, "h"'] \\ A \arrow[r, "g"] & B \end{tikzcd}$$

Prove that the following conditions on a ring $R$ with identity are equivalent:

(a) Every $R$-module is projective.

(b) Every short exact sequence of $R$-modules splits.

## III. Commutative algebra.

**1.** A non-empty subset $S$ of a ring $R$ with identity is called multiplicative if $ab \in S$ whenever $a, b \in S$.

(a) If $I$ is a proper ideal of $R$ disjoint from $S$, then prove that there exists an ideal $P$ maximal with respect to containing $I$ and being disjoint from $S$. Furthermore, show that $P$ is a prime ideal.

(b) If $P$ is a prime ideal of ring $R$, define the localization of $R$ at $P$ and prove that the localization has a unique maximal ideal.

**2.** Find a reduced primary decomposition for the following ideals. Also give the prime ideals to which the primary ideals belong.

(a) $(24)$ as an ideal in $\mathbb{Z}$.

(b) $(2x, x^2)$ as an ideal in $\mathbb{Z}[x]$.

(c) What theorems allow us to conclude that every ideal in $\mathbb{Z}[x]$ has a primary decomposition.

**3.** Let $R$ be a unique factorization domain and $K$ its field of fractions.

(a) Show that if $a \in K$ is integral over $R$, then $a \in R$. (In other words, $R$ is integrally closed.)

(b) True or false: If $S$ is a subring of $K$ containing $R$, then $S$ is also a unique factorization domain. If true, prove your answer. If false give examples of $R$, $K$, and $S$ that show it not to be true.

**4.** Determine, up to isomorphism, all semisimple rings of order $1008 = 2^5 \cdot 3^2 \cdot 7$.